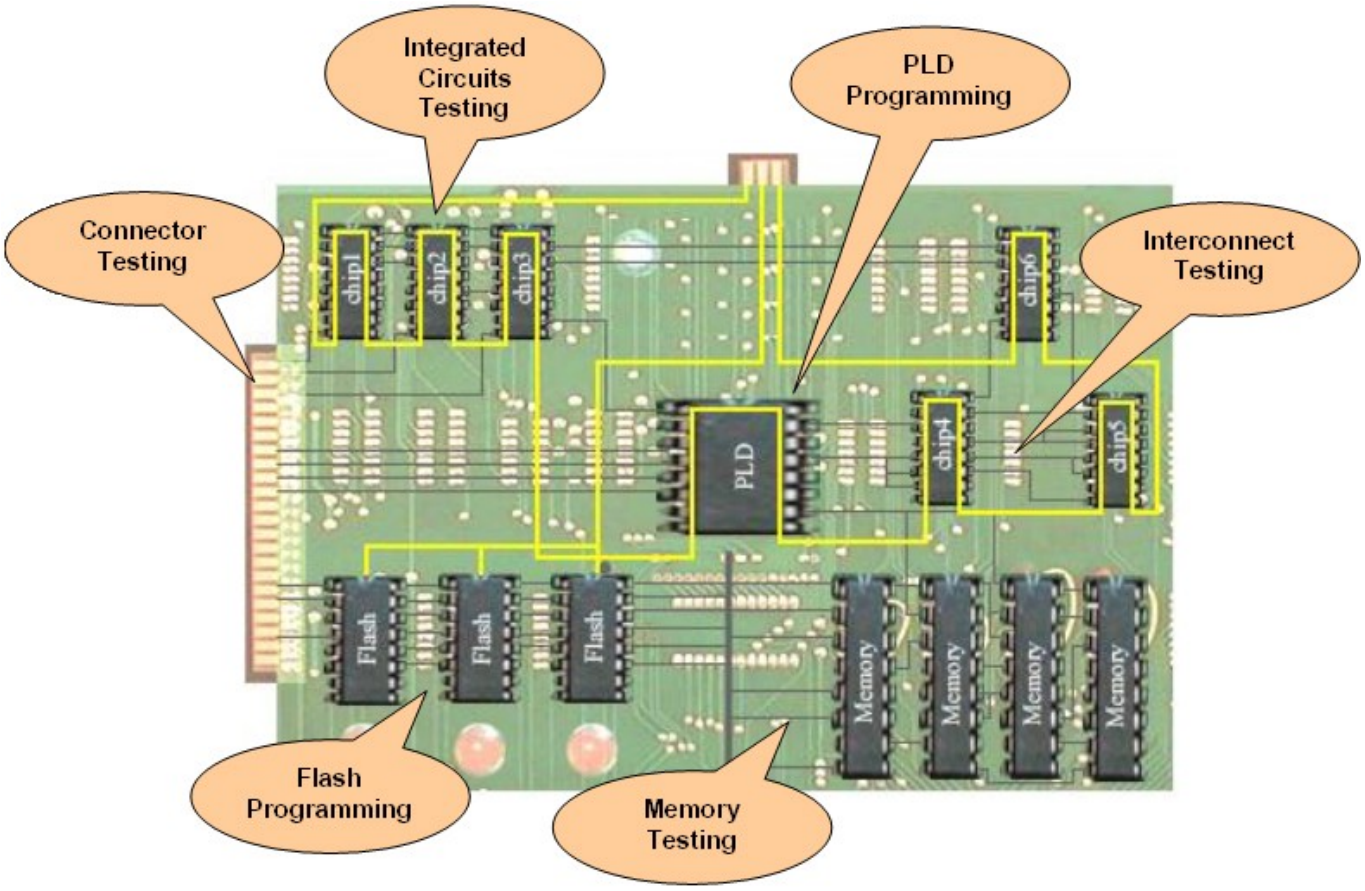
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Boundary-Scan Tests for ICs and PCB Assemblies



Amos Kingatua Follow

Dec 9, 2019 · 8 min read



Boundary-scan usage: Image

When it comes to testing a circuit assembly, you can test with simulation and you can test in-circuit. A boundary scan test is something else entirely, allowing you to test the state of individual traces while the circuit is active.

A boundary-scan test is a technique of checking ICs and interconnects on PCBs. It follows a testing architecture and procedures defined by the Joint Test Action Group (JTAG) under the IEEE 1149.1 standard.

The bounda ards where
physical acc stead of
probing of e rface to
send and receive specific test signals into the integrated circuits and boards under test.

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Unlike other testing methods that require expensive test equipment, the boundary scan technique relies on logic and uses minimal testing tools and is less costly and more effective. Also, the technique does not require a lot of physical access and will perform comprehensive testing as long as the circuit is designed according to the JTAG standard.

This test methodology, which was developed in the 80s for the Intel 80486 microprocessor, is today a standard technique that manufacturers and professionals use to program, debug and test almost all embedded devices and systems. Most semiconductor manufacturers are adding the test logic required to verify the chip functionality into the IC itself. This makes it easier to check and find defects in complex integrated circuits such as the FPGAs, CPLDs, DSPs and others without using the physical test probes.

Limitation of common test methods on complex and densely populated circuits

Today, the PCB market is shifting towards smaller footprints, high component densities, complex components, more functionalities, and shorter product life cycles. Manufacturers are increasingly using multilayer boards, ball-grid arrays (BGAs), surface-mount technology (SMT), multi-chip modules (MCMs), systems-in-package (SIPs), smaller PCB traces, increased IC pin-count, fine pitch components, and other technologies. All of these make it significantly more difficult to test the circuit during operation.

Despite the increased functions and complexity, manufacturers are under increased pleasure to release products faster to survive in the competitive world. With such trends and demands, PCB development and testing have become a challenge.

The above technologies make it difficult to access the pins hence limiting the effectiveness of the in-circuit tests (ICT), automatic test equipment (ATE) testing and other techniques that rely on mechanical probing. In particular, lack of direct access to pins, solder joints, and test points makes it difficult to perform some functional tests on the large and complex ICs and PCBs.

Although th ination of techniques t et.

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Out of the main test methods, the JTAG boundary-scan tests provide the best coverage of the PCOLA/SOQ/FAM defect-spectrum. Consequently, it reduces the need for additional structural testing methods and may even eliminate them depending on the PCB.

PCB Testing Area	Specific Defect Evaluated		Boundary-scan
Structural Devices	P	Presence	Yes
	C	Correctness	Yes
	O	Orientation	-
	L	Live	Yes
	A	Alignment	-
Structural Connections	S	Shorts	Yes
	O	Opens	Yes
	Q	Quality	-
Functional Connections	F	Features	Yes
	A	At-Speed	Yes
	M	Measurement	-

Boundary-scan test coverage of the PCOLA/SOQ/FAM defect spectrum: Image Electronicdesign

Boundary-scan architecture and how it works

The IEEE-1149.1 standard defines the boundary scan logic of JTAG compliant ICs. A typical JTAG ready IC comprises of boundary-scan cells, also known as logic cells placed between the chip's logic system and the signal pins or balls. Each cell has its specific role — most are either input or output only, but some are bidirectional.

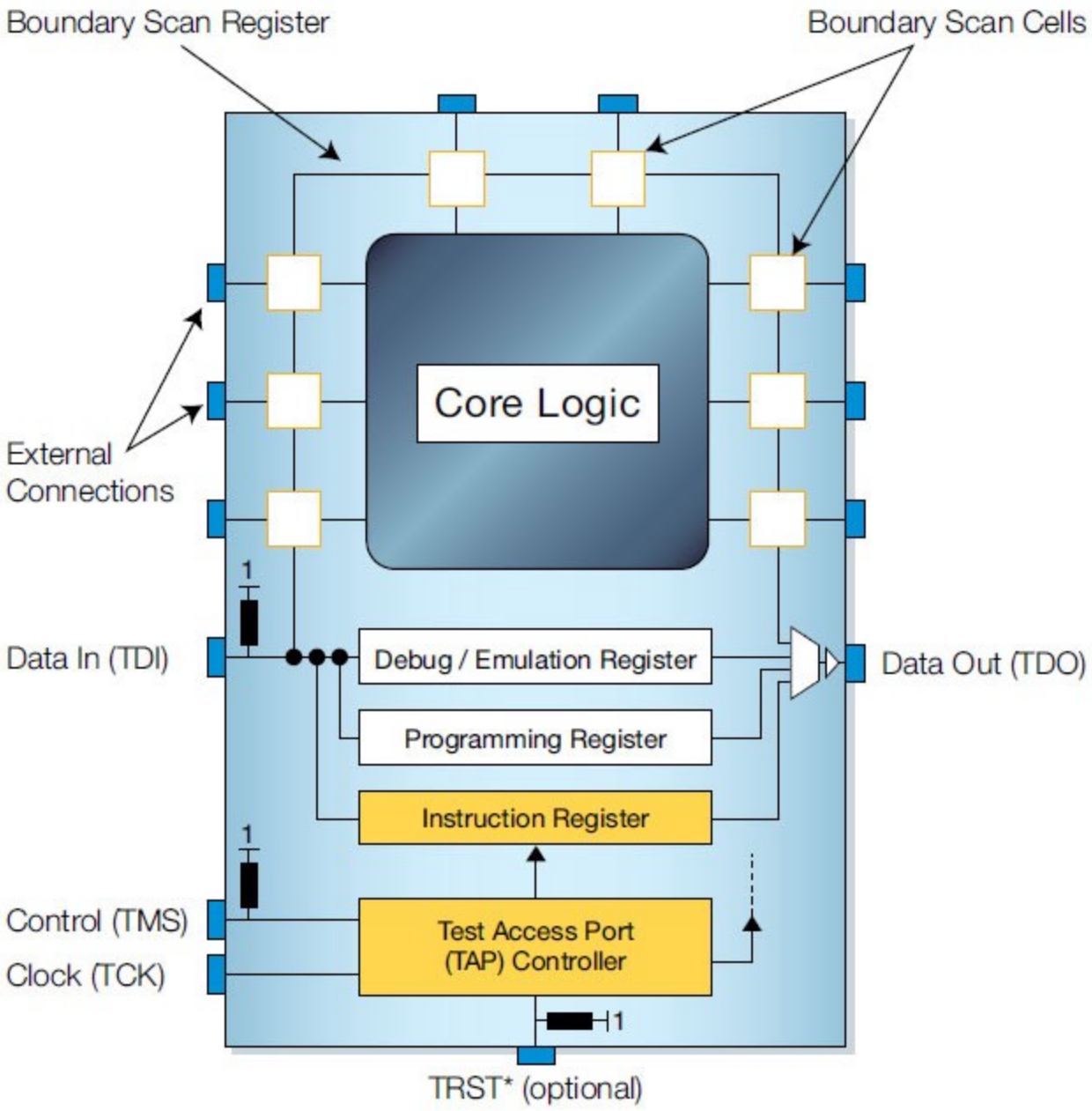
Usually, the cells are added at the chip's boundary — between the IC's core logic and the I/O pins or boundary hence the name boundary- scan. This establishes a serial test data path running through the entire IC. While in test mode, the cells control the status of the output pin while reading that of the input pin, hence enabling the testing of the IC and board interconnections. The cells usually appear transparent during the normal mode.

For a component to comply with the boundary scan JTAG standard, it must include;

1. A boundary-scan cell for each I/O pin.
2. The scan path where all boundary cells are serially connected
3. A four or five-wire Test Access Port (TAP) interface and controller to handle the boundary scan signals. The fifth wire is optional to provide for the reset signal).
4. Four or five additional pins for the JTAG signals.

5. Externa he
compon
informa process,
instructions, scan-cells available, design warnings, etc.

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A basic boundary-scan compliant IC: Image Xjtag

Test Access Port (TAP) Controller

A JTAG compliant IC has a Test Access Port (TAP) controller that comprises of the four signals and the logic that connects and controls them. The JTAG chip has four mandatory wires or pins and an optional fifth one for a reset signal. The required pins and roles are;

- TDI- serial input pin for the instructions, test, and programming data

- TDO- se

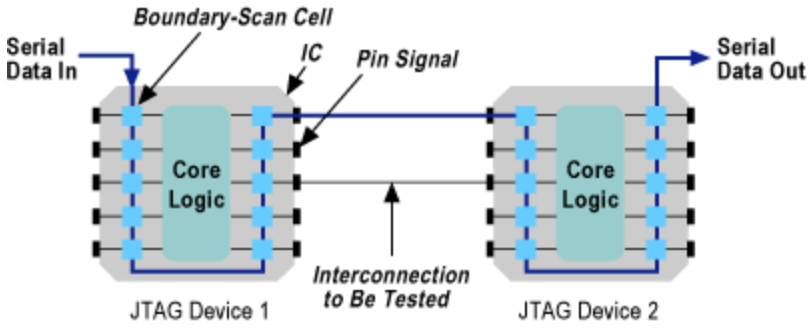
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- TMS- in
- TCK — clock signal input pin for the boundary scan circuitry
- TRST- reset signal (optional)

The TAP controller comprises of a 16-state finite state machine. These states are controlled by the test clock (TCK) and test modes select (TMS) signals. The JTAG interface provides a means to connect the external tools to the inbuilt TAP controller.

How Boundary scan testing works

In a typical boundary scan test, the tester sends diagnostic signals to the device's TDI input pin. The boundary-scan cells captures the signals and serially shifts them through the core logic which also scans them. The output is then serially shifted out of the core through the TDO pin.

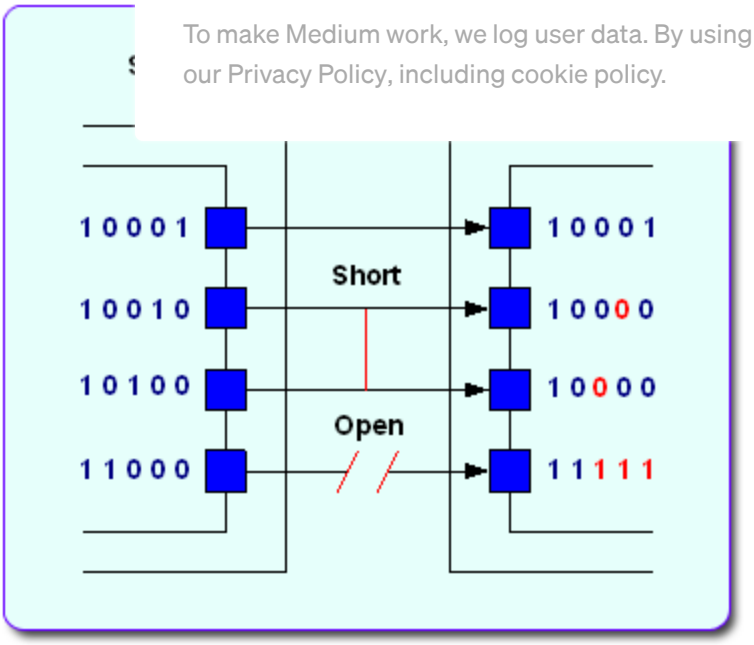


IEEE Std. 1149.1 Boundary-Scan Testing: Image Intel

The last step involves comparing the output with the expected result and consequently identifying if there are the shorts, opens, missing devices, dead components, internal defects within components, etc.

Usually, the boundary scan technique enables engineers to configure the cells in two main testing modes. This includes the internal testing mode to check the logic inside the chip while the external mode tests the interconnects between the ICs on the PCB.

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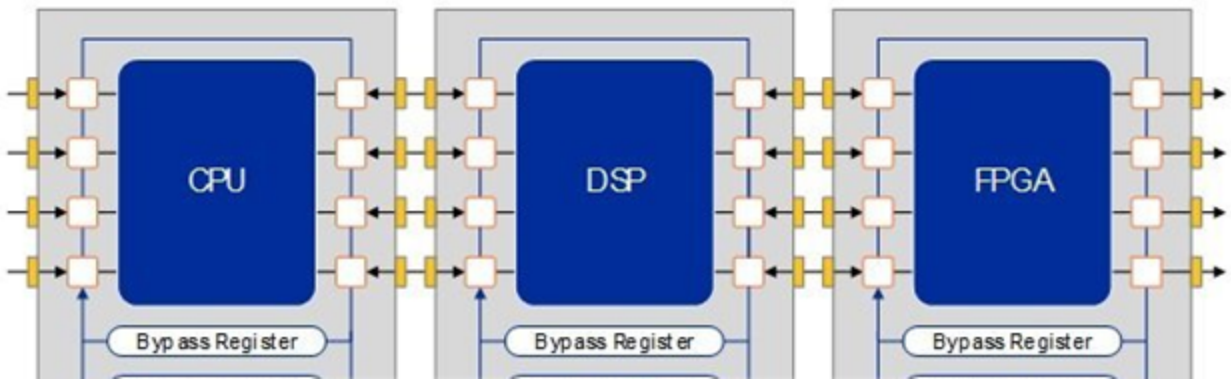
Interconnect test example: image Coleris

Designing a PCB for Boundary Scan Test

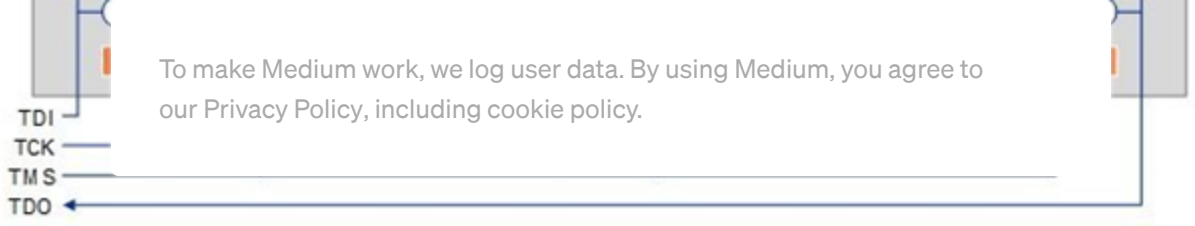
Besides the internal testing of ICs, designers can add boundary scan capability on the board level. Designing for boundary-scan tests has many benefits, especially for boards used in harsh environments, equipment that requires field repair, when there is a need to identify failure points to address for redesign purposes, etc. Generally, it is a good strategy when designing for testability.

Such a design begins by selecting IEEE 1149.1 or JTAG compliant devices and placing a test bus connector at an accessible location. For example, for card PCB, a good strategy is to route the test bus signals to the edge connector and assign them to spare pins.

In a PCB that uses many compliant devices, these are daisy-chained together. The output from one chip becomes the input to the next, and so on. The test signal is applied to the first IC on the chain while the output appears on the last device. Such an arrangement helps to verify the continuity of the connections between the various components' pins.



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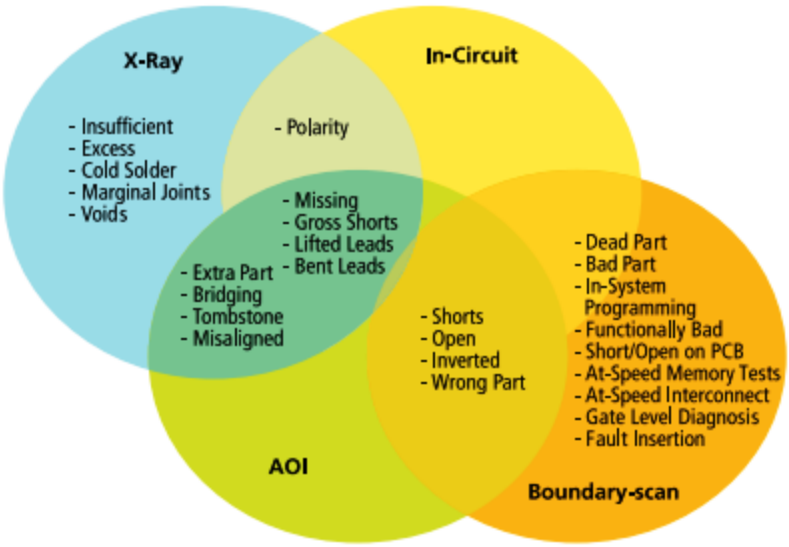
JTAG testing of multiple devices: Image Corelis

The boundary scan logic has a BYPASS mode that testers can use to shorten the chain and test time. Also, a dedicated JTAG bridge provides the ability to selectively configure the scan chain with multiple assemblies or components.

Benefits of boundary-scan testing

Benefits of the boundary scan include, but not limited to;

- Ability to test ICs and PCBs and with limited or no access to the internal connections and pins.
- Reducing the PCB testing efforts, costs and time without compromising on quality. Also, it offers reusable test patterns, better test coverage, and shorter time-to-market.
- Provides a low cost debugging and in-circuit programming of the CPLDs, serial EEPROMs, Flash, on-chip memory, etc.
- Reduces the risk of physically damaging the PCB, pins or creating shorts with the mechanical probes.
- Integrating the JTAG cells at the board level improves production and field testing while eliminating the need for other costly testing procedures and equipment.



Drawbac

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Despite the as requiring additional silicon for the boundary scan circuitry, extra pins, and more power consumption. Also, it requires more design efforts and devices may suffer from performance degradation.

While a boundary scan is useful in testing or reprogramming ICs and PCBs, criminals can use it to hack connected devices such as the IoT. Criminals can launch physical attacks by using the JTAG interface as a backdoor entry to access and exploit the devices. Consequently, the criminals can compromise data, cause a malfunction or stop the equipment.

Boundary-scan tools

With the increasing acceptance of the boundary scan as the main method for testing interconnects and performing in-system programming, various hardware and software companies have developed a wide range of tools. Today, there are several mature JTAG hardware and software tools in the market.

Typical boundary scan or JTAG tools are;

- Corelis — **ScanPlusTPG**
- Teradyne — **Victory**
- Acculogic — **ScanManager**
- Goepel Electronic — **System Cascon**
- Asset InterTech — **ScanWorks**
- Intellitech — **Eclipse Test Development Environment**
- Flynn Systems — **onTAP.**

Conclusion

Boundary-scan testing is a cost-effective and faster IC and PCB testing technique with wider coverage compared to other methods. The JTAG boundary-scan technique, which relies on industry-standard, IEEE 1149.1, provides a better, easy, and low-cost testing solution for PCBs with limited access to components pins and board-level circuitry.

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of today's cc cycle of the
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significant savings in product development, support time and costs.

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Pcb Testing Boundary Scan Test Jtag Test Interconnects In Circuit Test Probes

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